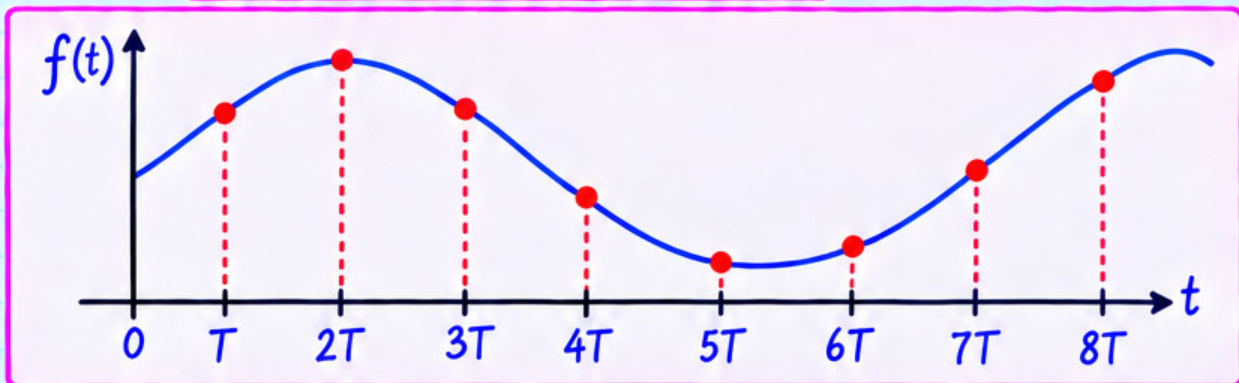


① Analog signal vs digital samples

- A continuous signal $f(t)$ has a value at every time t .
- Sampling keeps values at times $t = nT$.

$$x[n] = f(nT), \quad f_s = 1/T$$



② Sampling theorem

- If a signal contains no frequencies at or above B Hz, choose sampling rate $f_s > 2B$.
- This keeps repeated spectra from overlapping.

Example:

For $B = 1$ kHz, use $f_s > 2$ kHz.

Key idea:

- Sampling creates repeated copies of the signal spectrum at multiples of f_s .
- If $f_s > 2B$, the copies do not overlap.

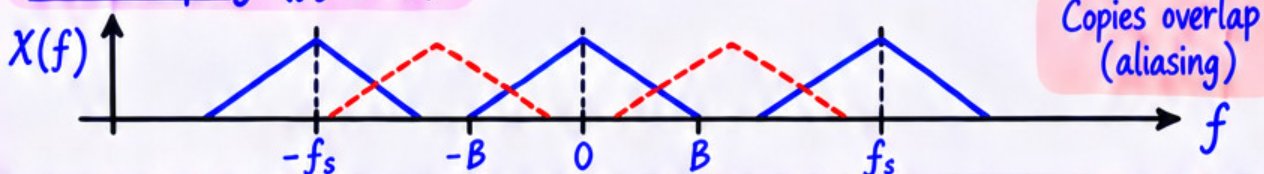
③ Aliasing and prevention

- When f_s is too low, different continuous frequencies can give the same samples.

Sufficient sampling ($f_s > 2B$)



Undersampling ($f_s \leq 2B$)



Key takeaway:

- Low-pass filter before sampling.
- Interpolation cannot reverse aliasing.

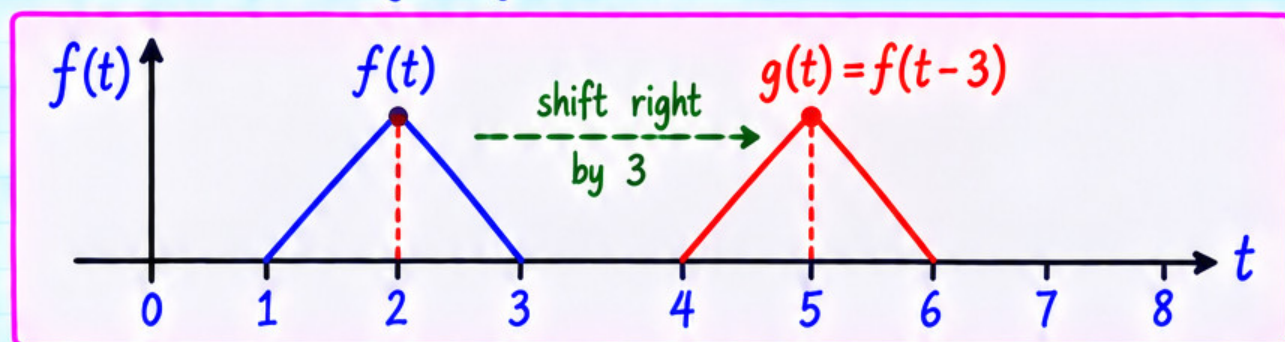
DIA: Shifting a waveform

Lecture extension - Fourier shift property

Amol Adhava
DIA - Page 2

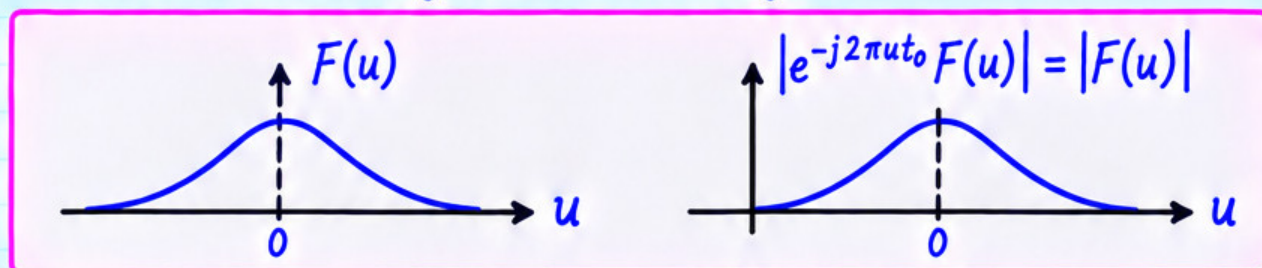
① Read the direction correctly

- $g(t) = f(t - t_0) \rightarrow$ move f to the RIGHT by t_0
- $g(t) = f(t + t_0) \rightarrow$ move f to the LEFT by t_0
- A pulse that peaked at $t=2$ now peaks at $t=5$ when shifted right by 3 seconds.



② Fourier-domain effect

- $f(t - t_0) \longleftrightarrow e^{-j2\pi ut_0} F(u)$
- The complex exponential changes phase. Its magnitude is 1, so the Fourier magnitude is unchanged.



③ Image translation

- Moving an image in x and y similarly adds a frequency-dependent phase factor.

$$f(x - x_0, y - y_0) \longleftrightarrow e^{-j2\pi(ux_0 + vy_0)} F(u, v)$$

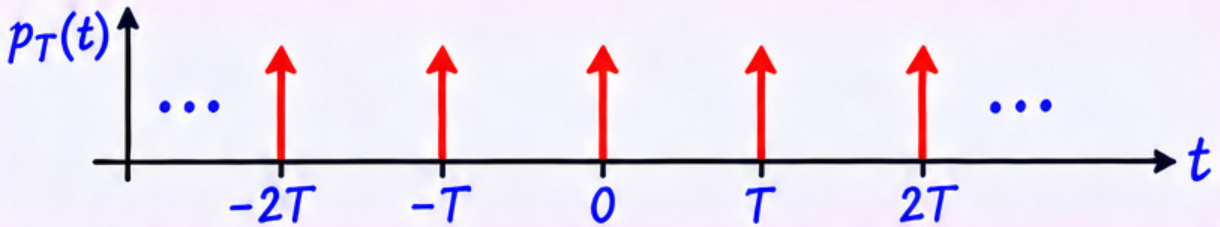
Key takeaway: ● Shift in position changes PHASE, not the magnitude spectrum (under the ideal transform model).

DIA: Periodic impulse train

① Sampling comb

- A periodic impulse train (sampling comb) is a train of Dirac impulses spaced by T .
- Each impulse is spaced by T ; sample frequency $f_s = 1/T$.

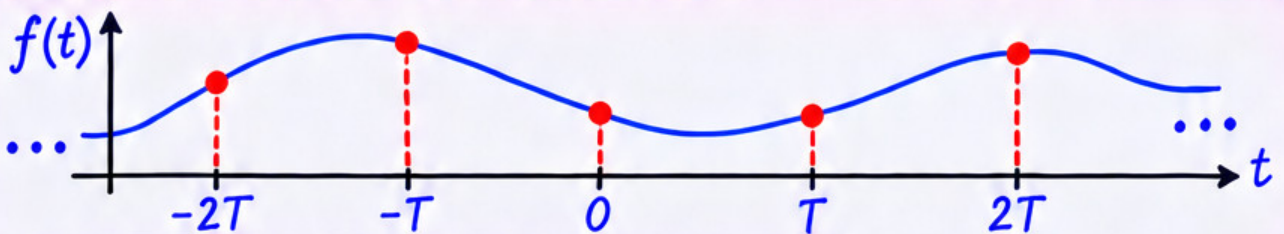
$$p_T(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT), \quad f_s = 1/T$$



② Sample the waveform

- Multiplying a continuous signal by the sampling comb gives weighted impulses at the sample instants.
- The heights of the impulses are the samples $f(nT)$.

$$f_{\text{samp}}(t) = f(t)p_T(t) = \sum_{n=-\infty}^{\infty} f(nT)\delta(t - nT)$$

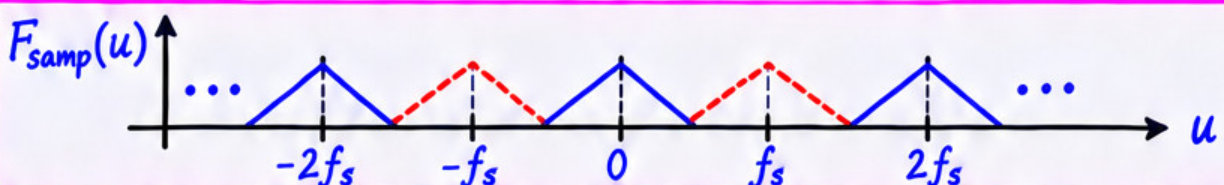


Example: For $T = 1 \text{ ms}$, $f_s = 1000 \text{ samples/s}$.

③ Frequency view

- The spectrum of the sampled signal is a periodic replication of the original spectrum $F(u)$ every f_s .
- Avoid aliasing when the spectral copies do not overlap.

$$F_{\text{samp}}(u) = \frac{1}{T} \sum_{k=-\infty}^{\infty} F\left(u - \frac{k}{T}\right)$$



Key takeaway: ● Smaller T increases f_s , spreading spectral copies farther apart.

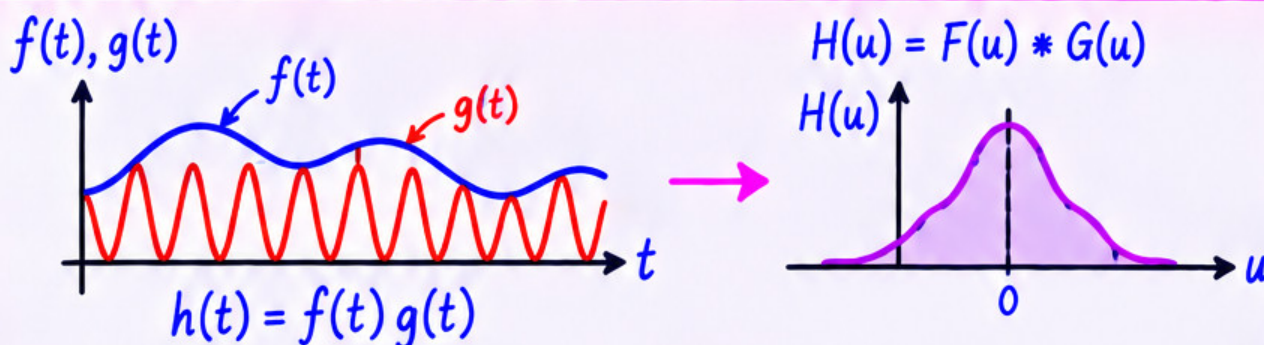
DIA: Multiplication of signals

Lecture extension - transform properties

① Multiply in time

$$h(t) = f(t) g(t) \leftrightarrow H(u) = F(u) * G(u)$$

- The star denotes convolution over frequency.
- Multiplication can spread or shift spectral content.

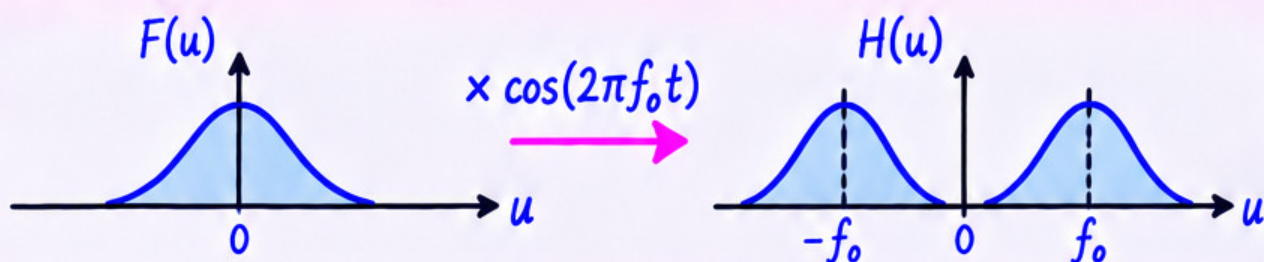


② A useful modulation example

$$g(t) = \cos(2\pi f_0 t)$$

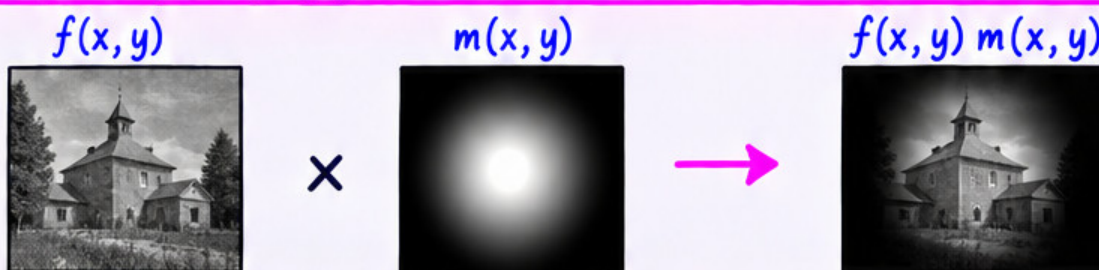
$$H(u) = \frac{1}{2} F(u - f_0) + \frac{1}{2} F(u + f_0)$$

- Multiplying by a cosine makes two shifted copies of the original spectrum.



③ Images use the same arithmetic differently

- Pixel-wise multiplication $f(x, y) m(x, y)$ is often used as a mask or spatially varying gain.



Key takeaway: ● Do not confuse a product in the spatial/time domain with a product of Fourier transforms.

DIA: Fourier transformation

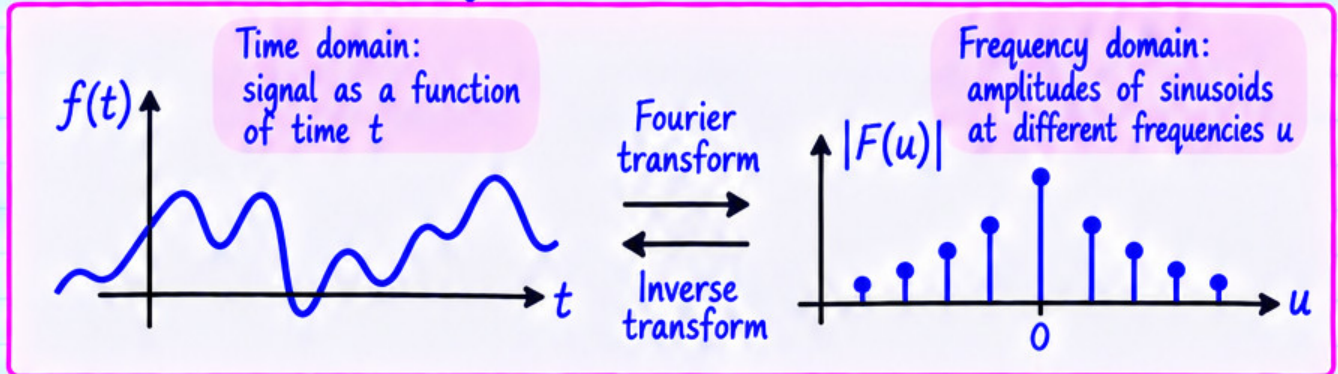
Ch. 2 introduces transforms; formal Fourier details in Ch. 4.

① What it tells us

- Fourier analysis rewrites a signal as a mixture of oscillations of different frequencies.

$$F(u) = \int_{-\infty}^{\infty} f(t) \exp(-j2\pi ut) dt$$

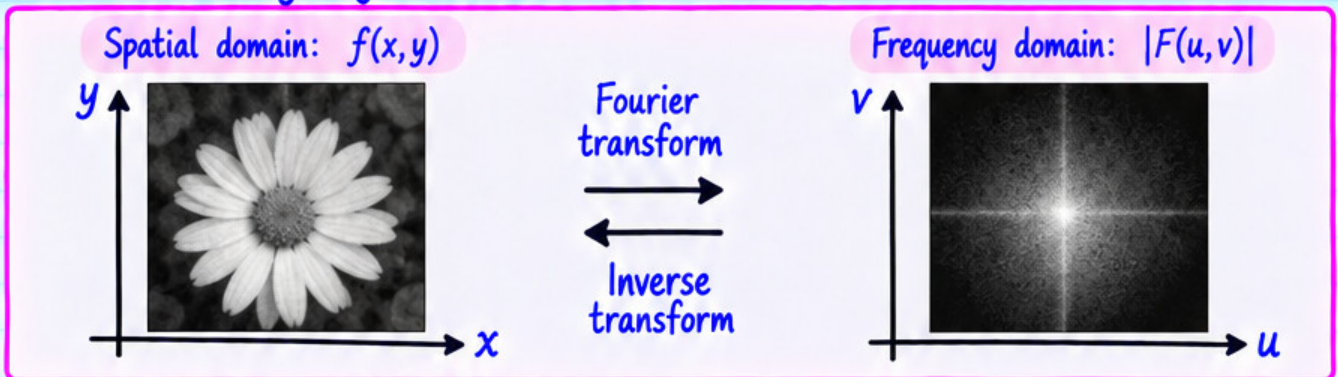
- The inverse transform combines those frequency components to reconstruct the signal.



② For an image

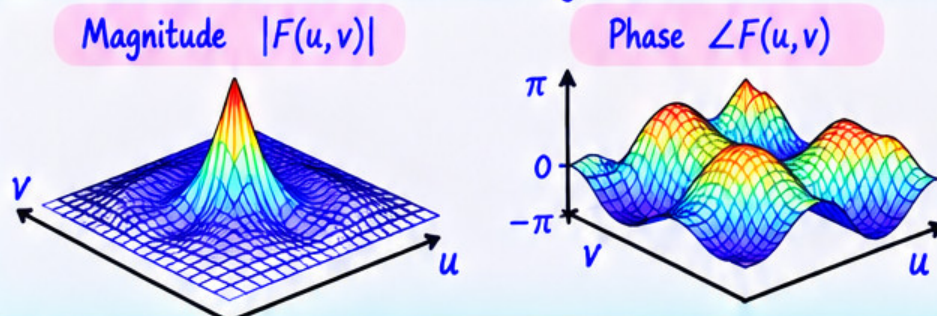
$$F(u, v) = \iint_{-\infty}^{\infty} f(x, y) \exp[-j2\pi(ux + vy)] dx dy$$

- Low spatial frequencies describe slow changes; high frequencies often carry edges and fine texture.



③ Magnitude and phase

- Magnitude says how strongly a frequency appears.
- Phase describes relative alignment and is vital for spatial structure.



Key takeaway:

- A shift changes Fourier phase; filtering changes component magnitudes and possibly phase.

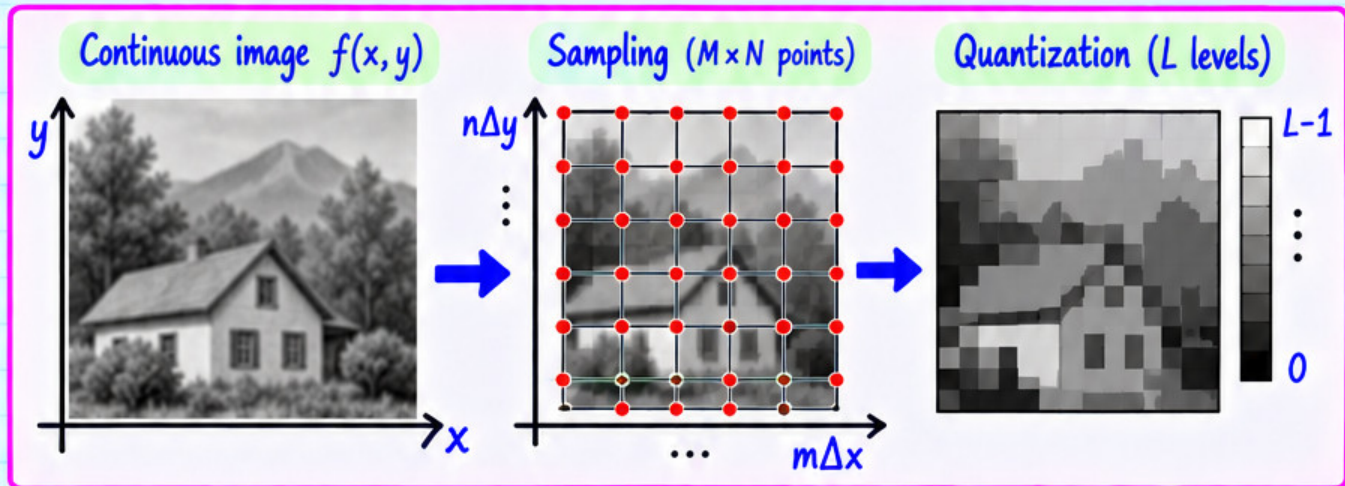
DIA: Sampling and Quantization

Chapter 2, section 2.4

① Two steps to a digital image

- Sampling chooses $M \times N$ spatial locations from a continuous image $f(x, y)$.
- Quantization rounds each sampled intensity to one of L allowed values.

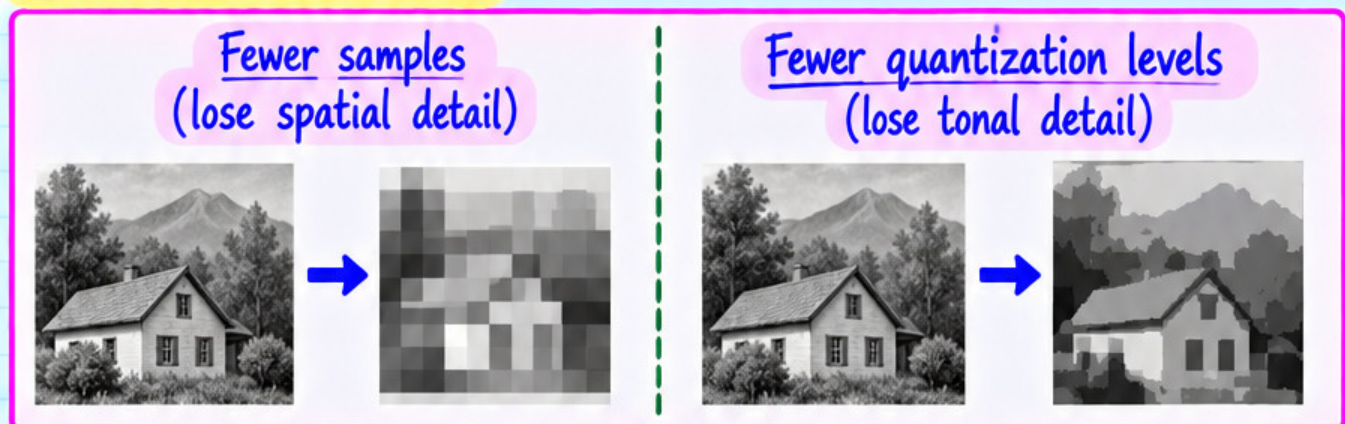
$$\text{digital pixel}[m, n] = Q(f(m\Delta x, n\Delta y))$$



② How many values?

- $L = 2^k$ levels for k bits per pixel
- An 8-bit grayscale image has 256 levels, usually labeled 0 to 255.
- raw storage = $M \times N \times k$ bits (ignoring file overhead)

③ Exam distinction



Key takeaway: ● Fewer samples lose spatial detail;
fewer quantization levels lose tonal detail.

DIA: Effect of bit resolution reduction

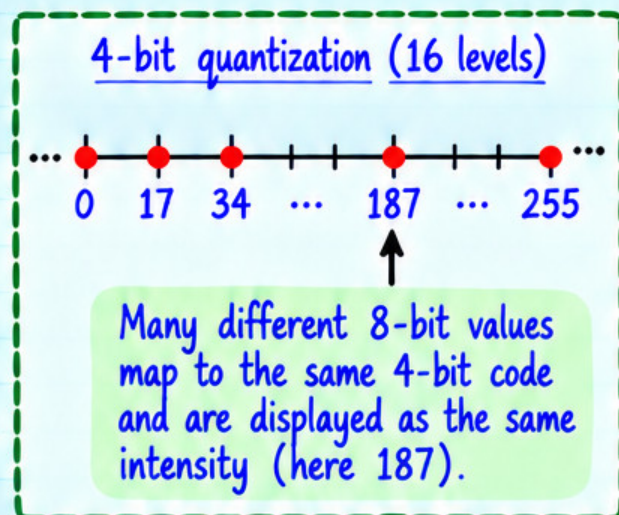
Chapter 2, section 2.4

① From 8 bits to k bits

- $q = \text{round}\left(\frac{(2^k - 1)r}{255}\right)$ for 8-bit r
- display estimate $\hat{r} = \text{round}\left(\frac{255q}{2^k - 1}\right)$
- With $k = 4$, only 16 encoded intensities remain instead of 256.

② Worked pixel

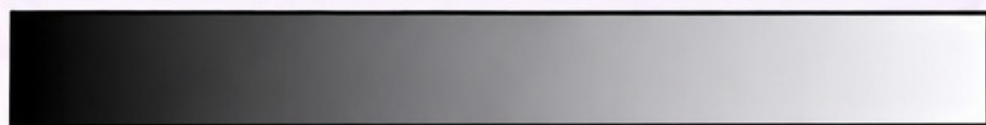
- For $r = 183$ and $k = 4$:
 $q = \text{round}(15 \times 183 / 255) = 11$
display estimate = 187.
- Many input values map to the same code 11, so their differences cannot be recovered.



③ Visual effect

- Smooth gradients become bands (posterization).
- Dithering can make bands less obvious by adding patterned noise.

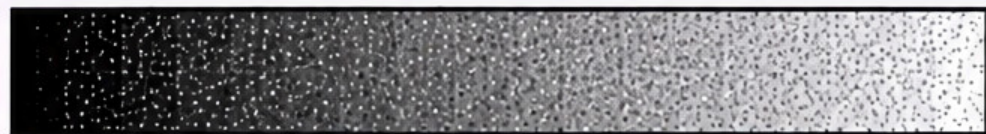
Original
8-bit gradient
(256 levels)



Reduced to 4 bits
(16 levels)



With dithering
(4 bits + noise)



Key takeaway:

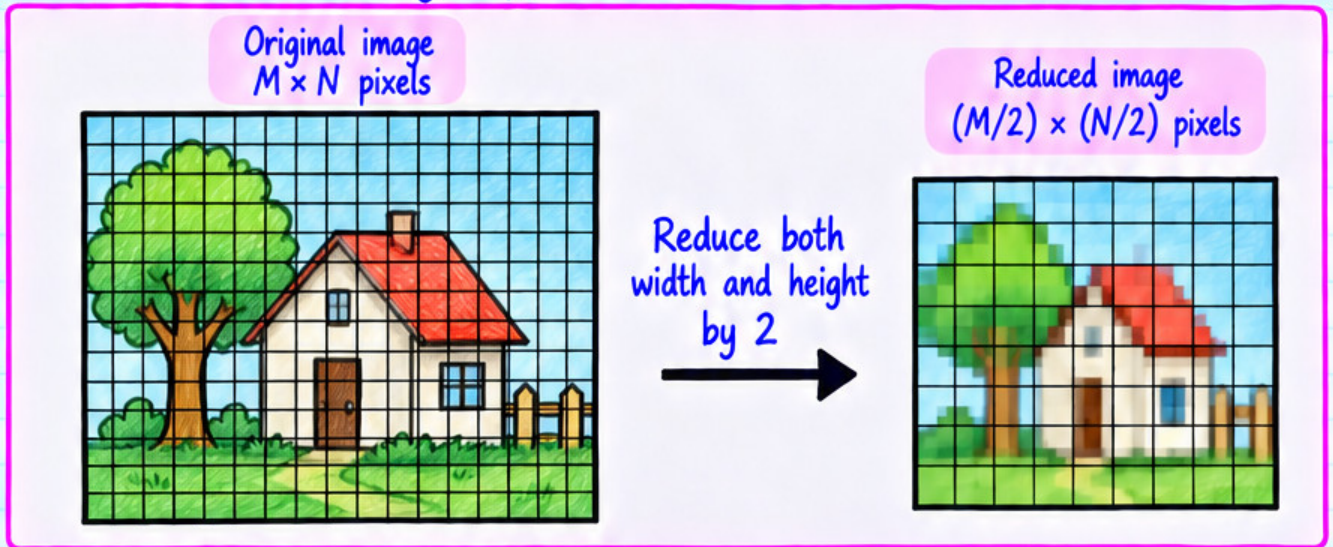
- Increasing display bit depth later does not restore the discarded original intensity information.

① Fewer spatial samples

- Reducing both width and height by half leaves one quarter of the original pixel count.

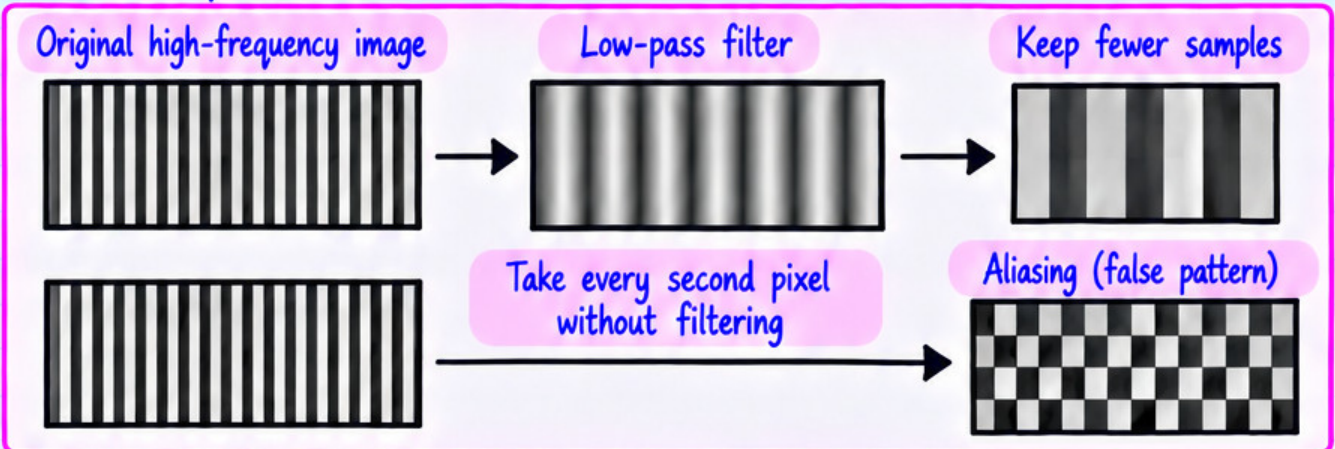
$$(M/2) \times (N/2) = MN/4$$

- Thin lines, small text and high-frequency textures can disappear or change appearance.



② Correct reduction workflow

- First low-pass filter to suppress detail above the new Nyquist limit; then keep fewer samples.
- Simply taking every second pixel may alias stripes into false patterns.



③ Separate from bit depth

Key takeaway:

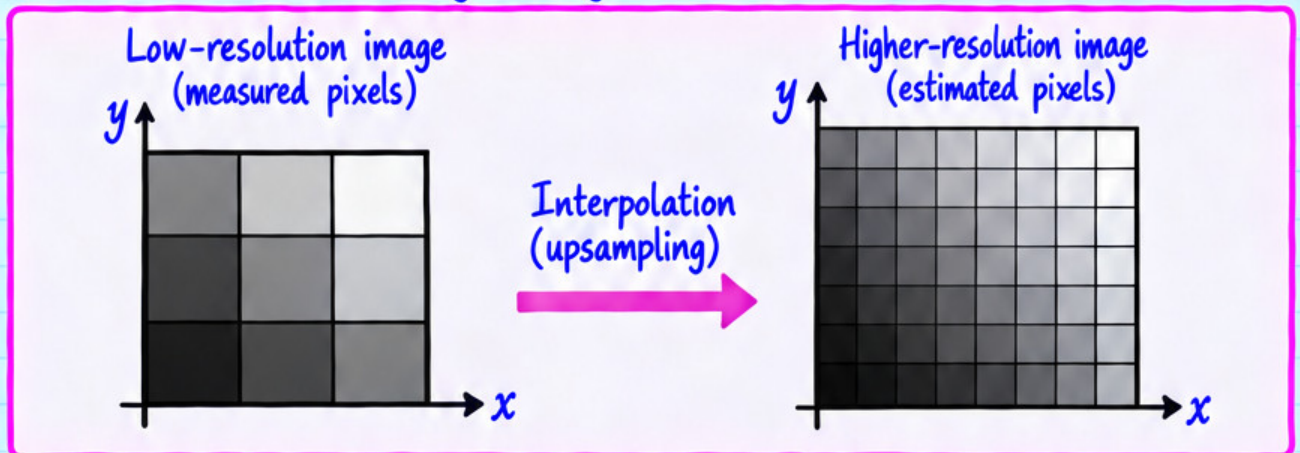
- Spatial reduction changes M and N .
- Bit-depth reduction changes L .
- They are not interchangeable.

DIA: Resolution Improvement

Chapter 2, section 2.4 — image interpolation

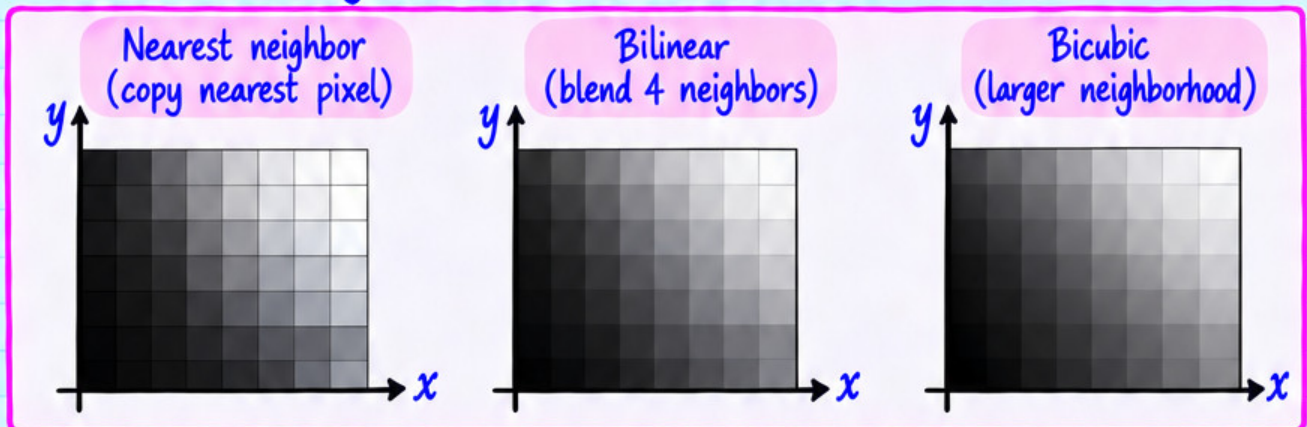
① What upsampling does

- Add a denser output grid and estimate intensities between measured pixels.
- Nearest neighbor copies; bilinear blends four neighbors; bicubic uses a larger neighborhood.



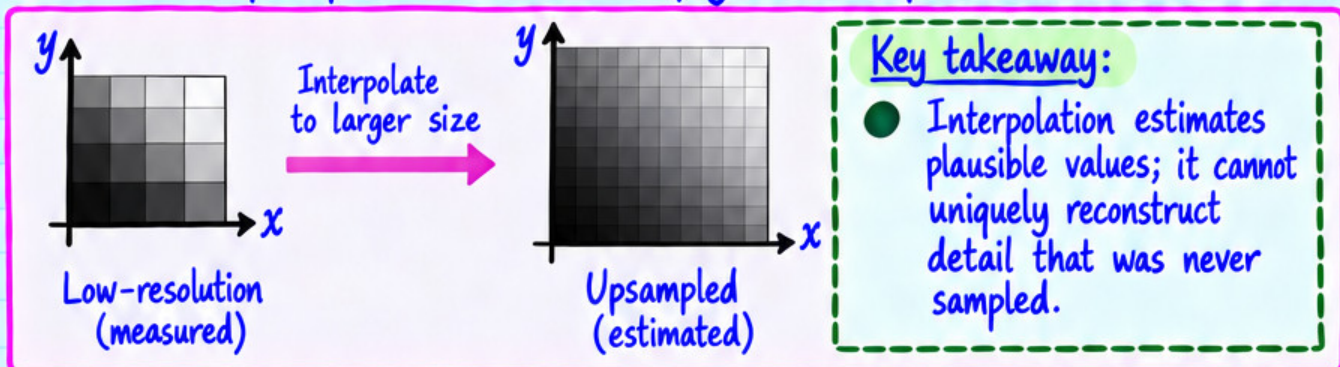
② Choose by purpose

- Nearest neighbor keeps class labels intact but looks blocky on photographs.
- Bilinear smooths but can blur edges. Bicubic can retain smoother edges, with more calculation.



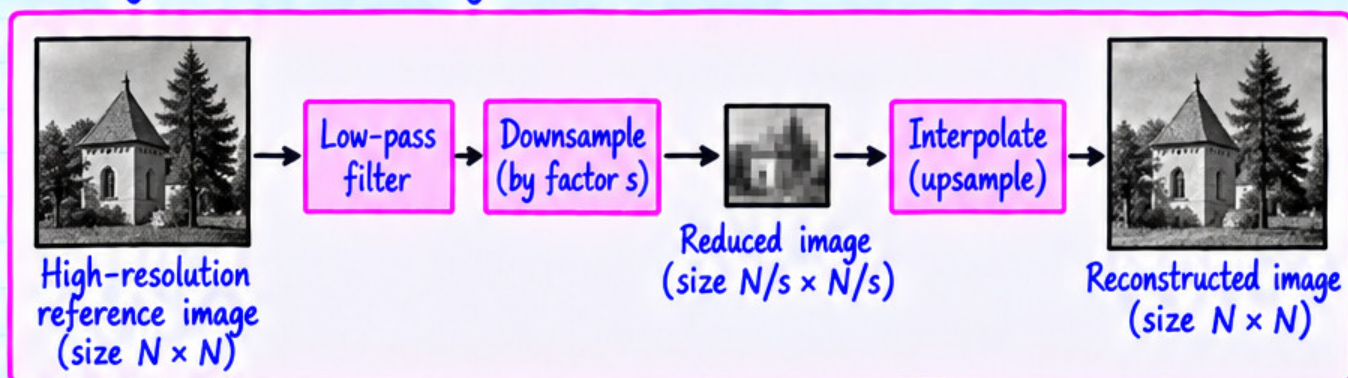
③ Limits

- More output pixels does not imply more captured scene detail.



① Controlled experiment

- Begin with a high-resolution reference image. Apply appropriate low-pass filtering, then downsample.
- Upsample the reduced image using each candidate algorithm to the original reference dimensions.



② Numerical comparisons

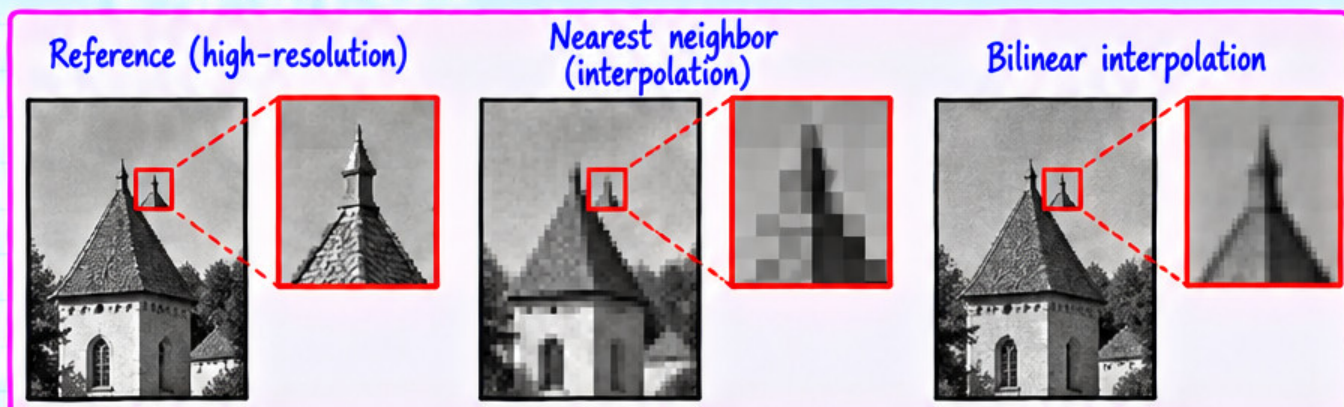
- $$MSE = \frac{1}{MN} \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} (r[m, n] - \hat{x}[m, n])^2$$

- $$PSNR = 10 \log_{10} \left(\frac{MAX^2}{MSE} \right) \text{ dB}$$

- For 8-bit images $MAX = 255$. Lower MSE and higher PSNR mean closer pixel values.

③ Visual comparisons

- Inspect edges, lettering, textures, ringing and blockiness at the same zoom.



- Key takeaway:
- A high PSNR is useful but may not match perceived sharpness.
 - Test several kinds of image.

DIA: Number of intensity levels

Chapter 2, section 2.4

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① What changes as L varies?

$$L = 2^k$$

$k = 8 \rightarrow 256$ levels

$k = 4 \rightarrow 16$ levels

$k = 2 \rightarrow 4$ levels

- At 256 levels a smooth gradient can look continuous; with 16 or 4 levels, steps are easier to see.

$L = 256$ ($k = 8$)



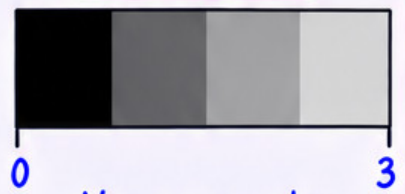
Looks smooth

$L = 16$ ($k = 4$)



Steps visible

$L = 4$ ($k = 2$)



Very coarse steps

② The histogram changes too

- Reducing L merges several source gray values into a smaller number of output bins.
- The effect depends on the picture: a nearly uniform region may show bands sooner than a textured one.

Original histogram (larger L)



After reducing L



③ Memory connection

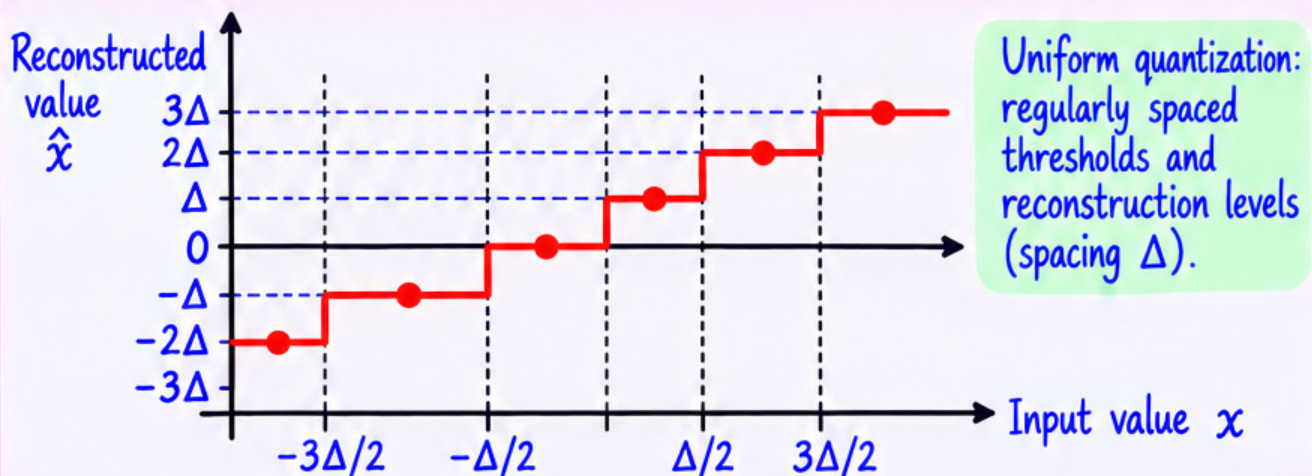
- Raw grayscale image of size $M \times N$ with k bits per pixel needs

$M \times N \times k$ bits for raw grayscale storage.

Key takeaway: More levels improve possible tonal precision, but cannot fix a spatially undersampled image.

① From measurement to code

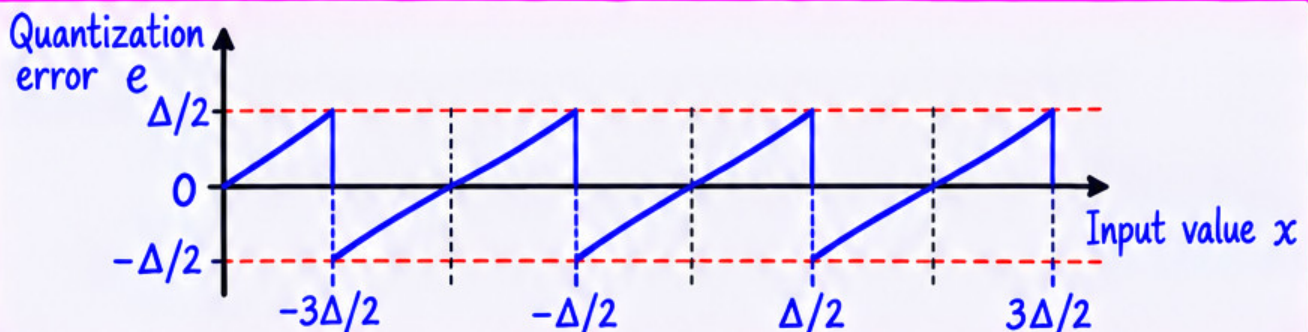
- Set an input range and L output codes. Every measured intensity is assigned to a code or reconstruction level.
- Uniform quantization uses regularly spaced thresholds; rounding chooses the closest reconstruction level.



② Quantization error

error e = original value - reconstructed value

- For a uniform step of width Δ , nearest-level rounding gives $|e| \leq \Delta/2$ away from overload.
- Clipping outside the supported input range creates overload error larger than the ordinary rounding error.



③ Exam takeaway

- Quantization maps many possible inputs to one code. It is lossy unless the inputs already lie on allowed levels.

Key takeaway:

- Many possible input values map to the same code.
- Quantization is lossy unless the inputs already lie exactly on allowed levels.

1 Decompose an 8-bit pixel

- An 8-bit pixel value r (0 to 255) can be written as a sum of its bit-plane contributions.

$$r = b_7(128) + b_6(64) + \dots + b_1(2) + b_0(1)$$

- Each bit b_i can be extracted using

$$b_i = \left\lfloor \frac{r}{2^i} \right\rfloor \bmod 2$$

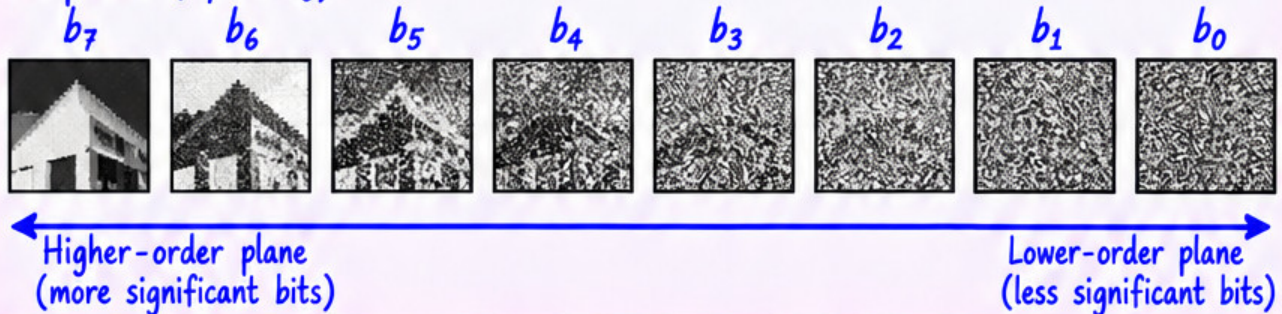
- For $r = 183$, the 8-bit binary representation is 10110111. The planes $b_7, b_6, b_5, b_4, b_3, b_2, b_1, b_0$ are:

$r = 183$ (8-bit: 10110111)	b_7	b_6	b_5	b_4	b_3	b_2	b_1	b_0
	1	0	1	1	0	1	1	1

2 Why inspect planes?

- High-order planes often retain prominent structure. Lower-order planes may contain fine texture or noise.
- Plane appearance depends on the image; a low plane is not always a meaningless noise.

Bit-planes ($b_7 \dots b_0$)



3 Reconstruction

- The original 8-bit pixel can be reconstructed by summing all bit-plane contributions.

$$r = b_7(128) + b_6(64) + \dots + b_1(2) + b_0(1)$$

Key takeaway: The exact original pixel is recovered only when all bit-plane contributions are included.

DIA: Image histogram

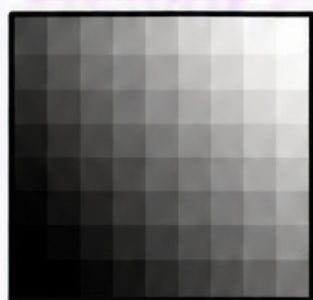
Chapter 3, section 3.3

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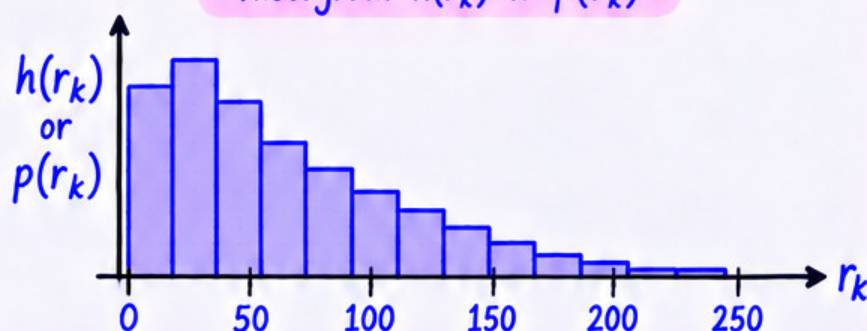
① Count intensities

- $h(r_k) = n_k$ pixels at intensity r_k
- $p(r_k) = \frac{n_k}{MN}$, so $\sum_k p(r_k) = 1$
- A histogram says how often each intensity occurs, not where those pixels are located.

Image $f(x,y)$



Histogram $h(r_k)$ or $p(r_k)$



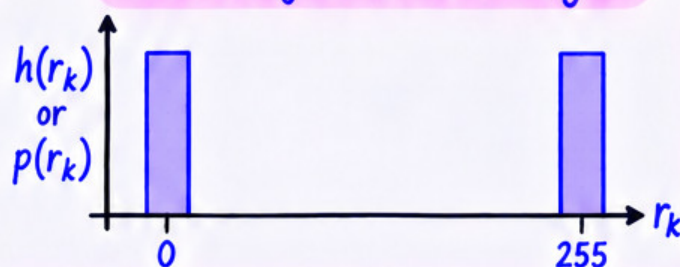
② How to read it

- Counts clustered at low intensities suggest a dark image. A narrow spread may indicate low global contrast.
- Two very different images can have the same histogram if their pixel values are rearranged.

Two images with the same histogram

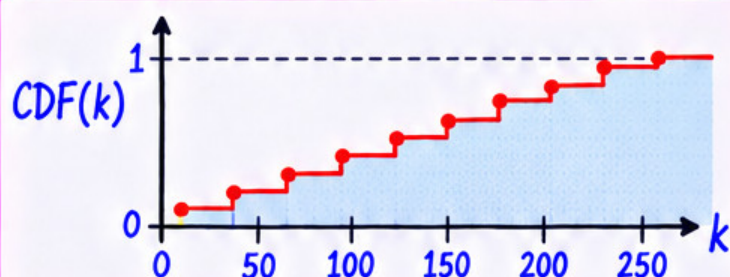


Same histogram for both images



③ Cumulative histogram

- $CDF(k) = \sum_{j=0}^k p(r_j)$



Key takeaway:

- The cumulative distribution never decreases and is the basis of histogram equalization.

DIA: Combining Images

Chapter 2, section 2.6

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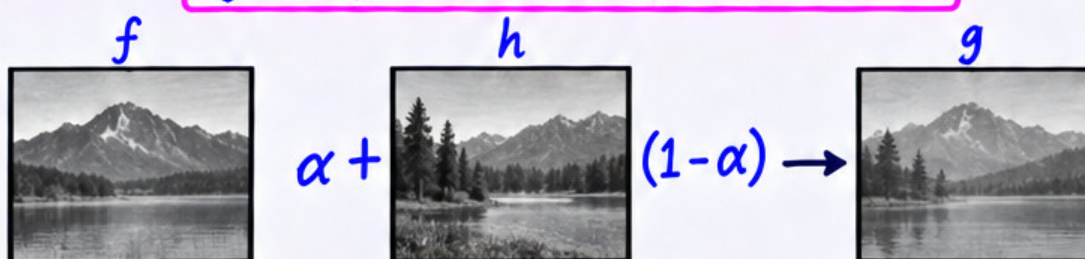
1 Prepare corresponding pixels

- Images must be aligned and have compatible dimensions for direct pixel-wise combination.
- Work in floating point or a wider integer type, then choose clipping or scaling for display.

2 Three common combinations

blend:

$$g = \alpha f + (1 - \alpha) h, \quad 0 \leq \alpha \leq 1$$



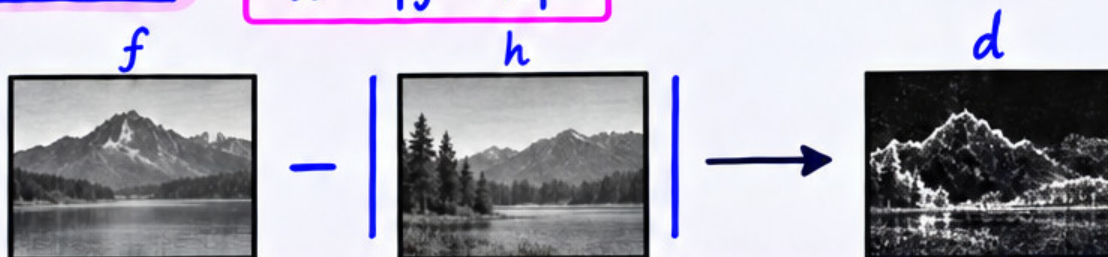
mask:

$$g = m f, \quad \text{with } m \in \{0, 1\}$$



difference:

$$d = |f - h|$$



- Blending merges views; masking selects a region; difference highlights change.

3 Caution

Key takeaway:

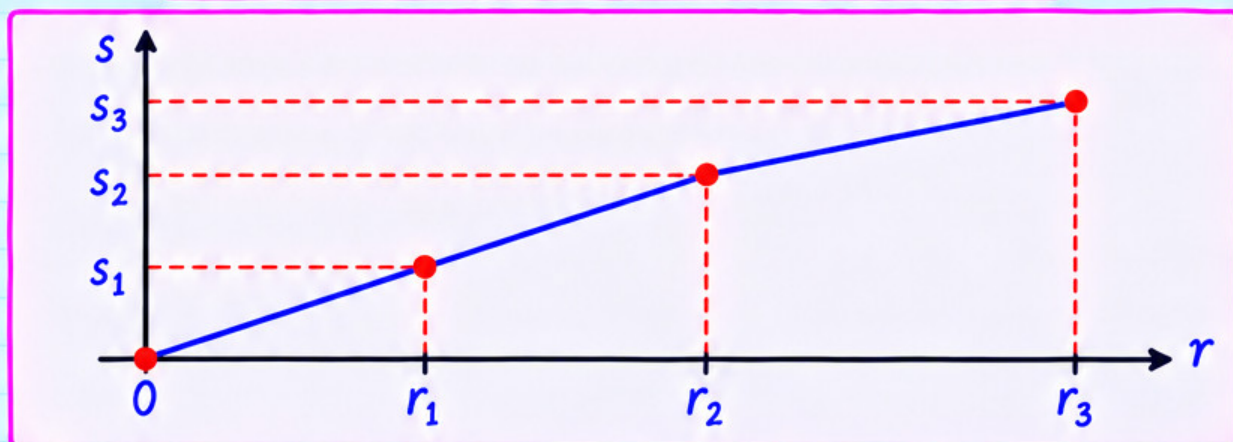
- Misregistration creates ghost edges.
- Repeated clipping destroys values beyond the output range.

① Control points determine the shape

- Choose input/output points (r_1, s_1) , (r_2, s_2) , then join adjacent points with straight segments.

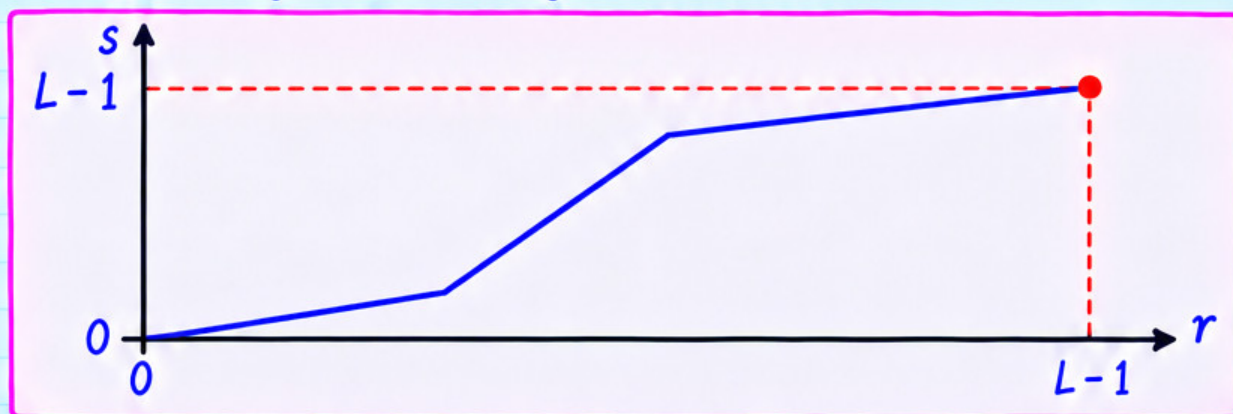
Within a segment:

$$s = s_1 + (r - r_1) \frac{s_2 - s_1}{r_2 - r_1}$$



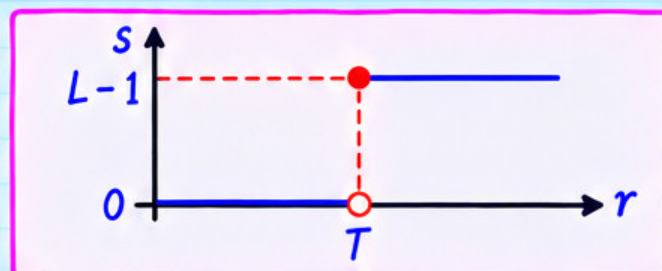
② Contrast stretching

- A steep segment expands contrast for intensities in that interval; flatter segments compress the rest.
- Intensity slicing makes a chosen band prominent while suppressing or preserving other values.



③ Special case: threshold

$$s = \begin{cases} 0 & \text{when } r < T \\ L-1 & \text{when } r \geq T \end{cases}$$



Key takeaway: ● A monotonic transform preserves ordering, but flat segments and clipping merge distinct inputs.

DIA: Histogram Equalization: idea

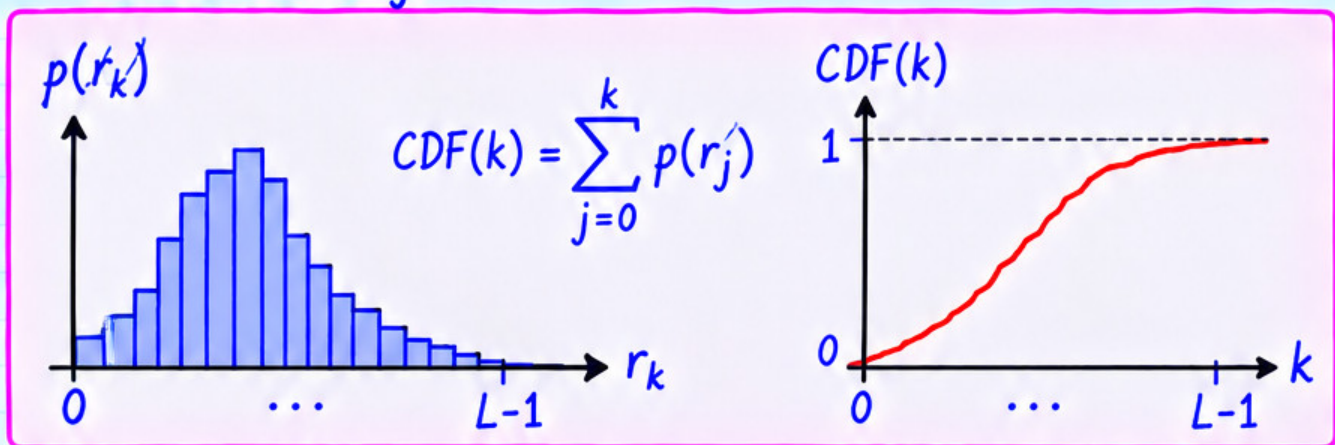
Chapter 3, section 3.3, begins at p. 148

① Build one monotonic lookup table

- Count intensities, normalize counts to $p(r_k)$, and accumulate a CDF from dark to bright.

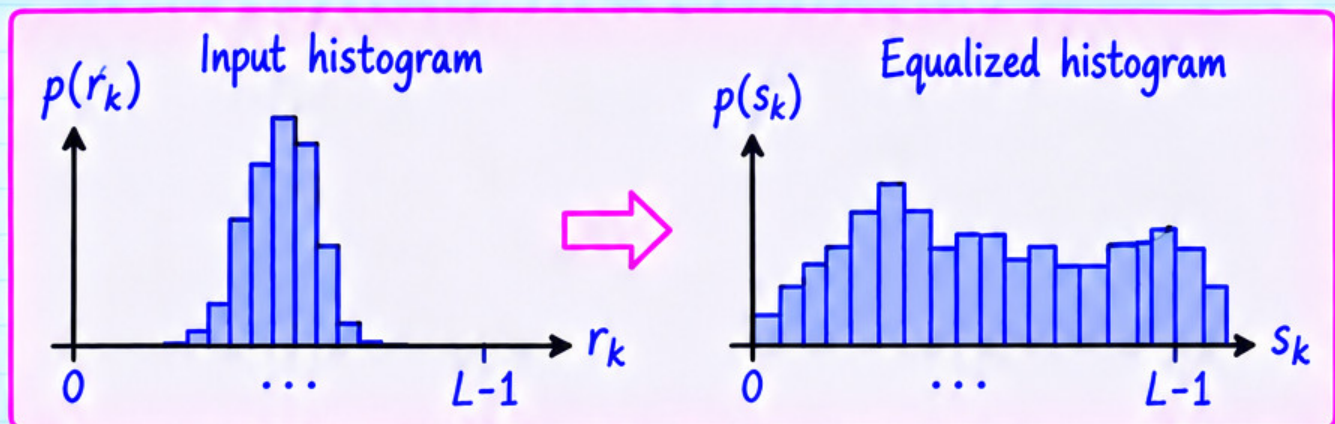
$$s_k = \text{round}((L-1) \times \text{CDF}(k))$$

- Map every input pixel using the lookup table for its old intensity.



② Expected effect

- A narrow range of intensities can be redistributed over a wider part of the output range.
- The result may improve global contrast, but can also make noise or unnatural tones more visible.



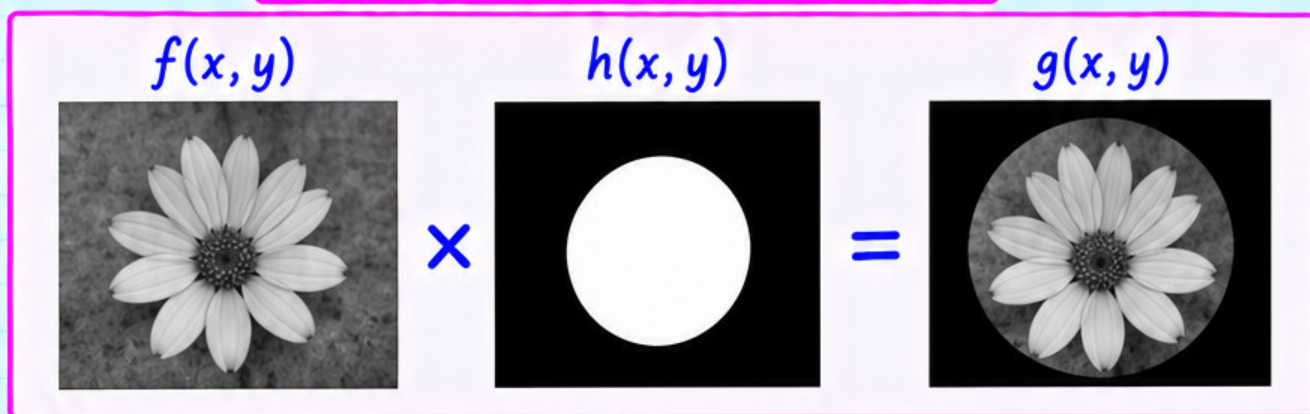
③ Discrete caveat

- Key takeaway:
- Equalization does not guarantee a perfectly flat output histogram: integer rounding and tied values remain.

1 Multiplication

- Use a 0/1 mask to retain selected pixels.
- Use a smoothly varying gain image to change illumination.

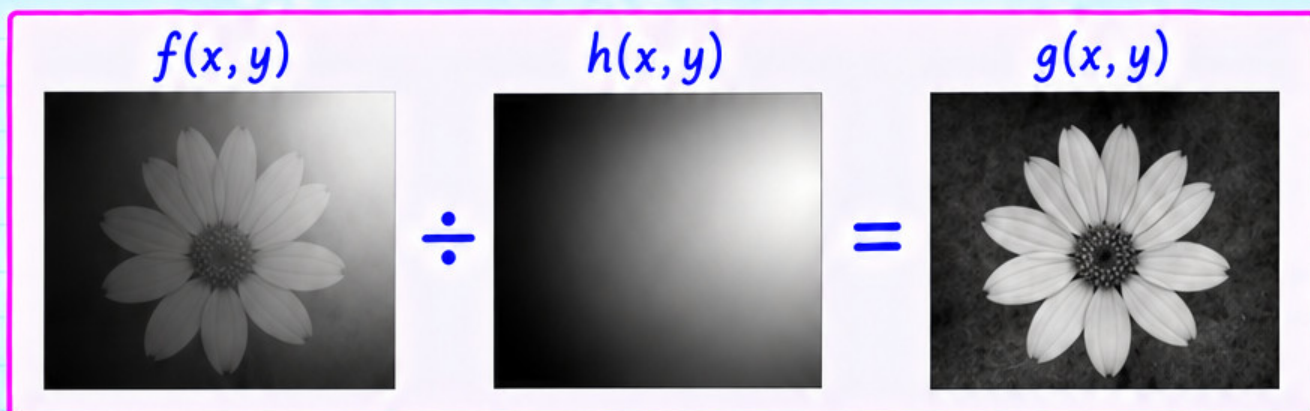
$$g(x, y) = f(x, y) \times h(x, y)$$



2 Division

- Division can compensate for a measured illumination field h when h is reliable.

$$g(x, y) = f(x, y) / (h(x, y) + \epsilon)$$



3 Numerical safety

- Normalize ranges as needed.
- A denominator near zero amplifies noise; ϵ prevents literal division by zero.

Key takeaway:

- Perform arithmetic in float/wider precision, then clip or rescale to the target image range.

DIA: Image Addition & Subtraction

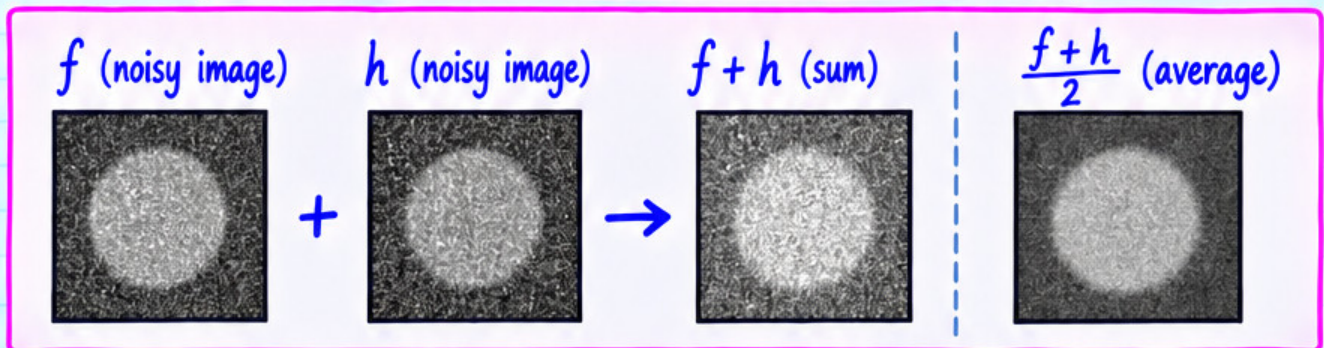
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Chapter 2, section 2.6

① Addition and averaging

- Averaging registered independent noisy images can reduce random, zero-mean noise.

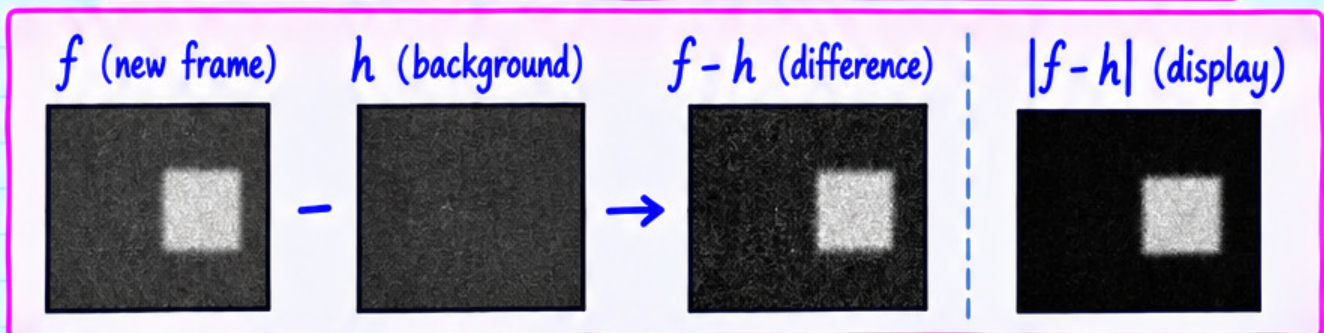
$$\text{sum} = f + h \quad \text{average} = \frac{f + h}{2}$$



② Subtraction and change

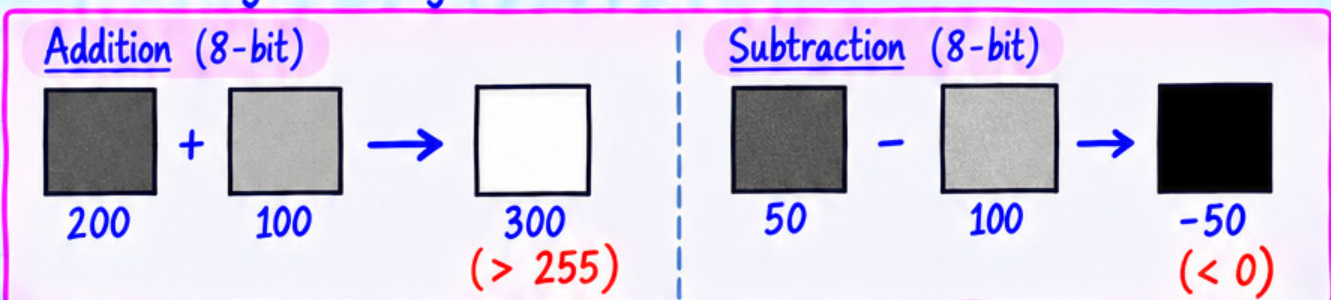
- A background model subtracted from a new frame can highlight moving or changed regions.

$$\text{difference} = f - h \quad \text{display often uses } |f - h|$$



③ Range matters

- 8-bit addition can exceed 255 and subtraction can go below 0; saturating too early loses information.



Key takeaway: ● Align images first and use signed or floating-point intermediate values before final display scaling.

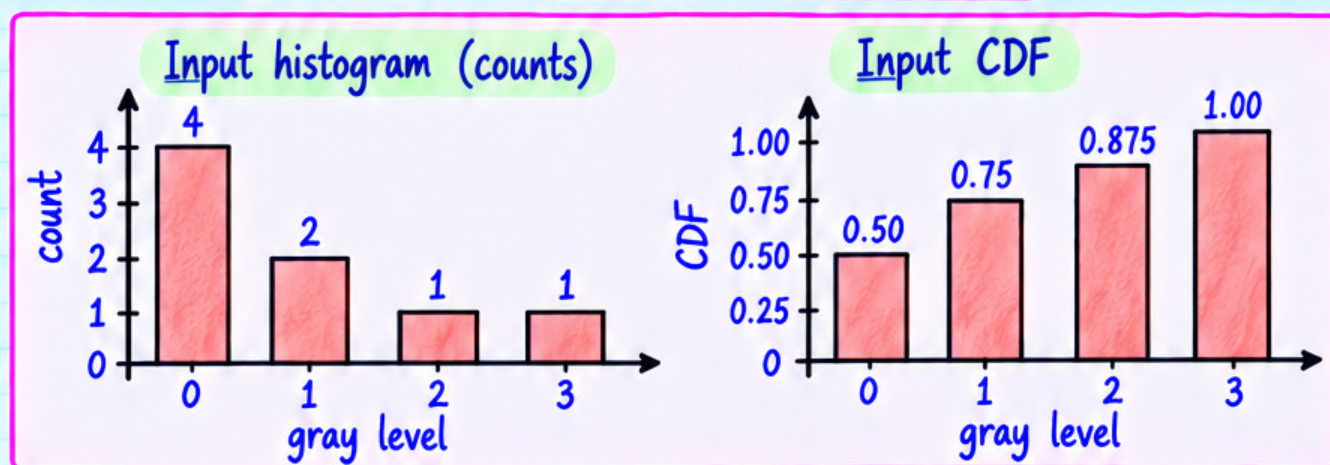
DIA: Histogram Equalization: example

Chapter 3, section 3.3 - worked discrete case

① Input four gray levels

- For levels 0, 1, 2, 3, suppose counts are [4, 2, 1, 1].
- Total pixels $N = 8$.

$$p = [0.50, 0.25, 0.125, 0.125]$$
$$\text{CDF} = [0.50, 0.75, 0.875, 1.00]$$

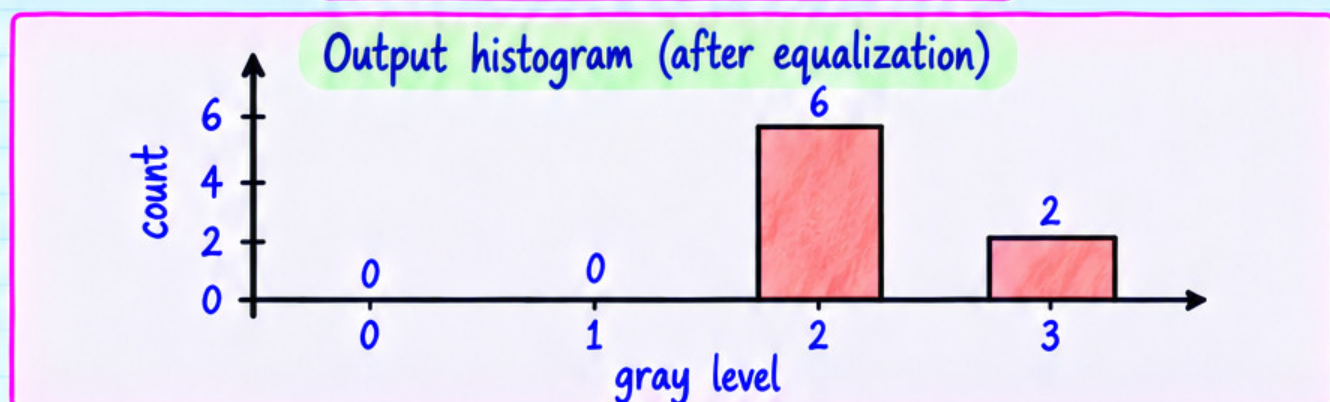


② Compute the output levels

$$s_k = \text{round}(3 \times \text{CDF}) \rightarrow [2, 2, 3, 3]$$

- Input levels 0 and 1 both become 2;
input levels 2 and 3 both become 3.

$$\text{output counts} = [0, 0, 6, 2]$$



③ Meaning

- Even with the equalization rule, the output is not uniform.

Key takeaway:

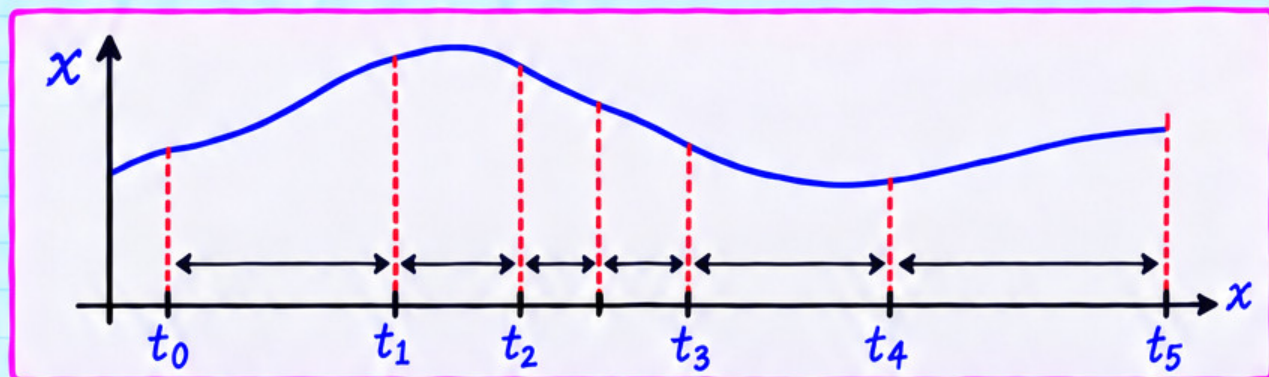
- Show every CDF and mapping step in an exam.
- The output histogram is not uniform.

DIA: Non-uniform Equalization

Clarification of two related lecture terms

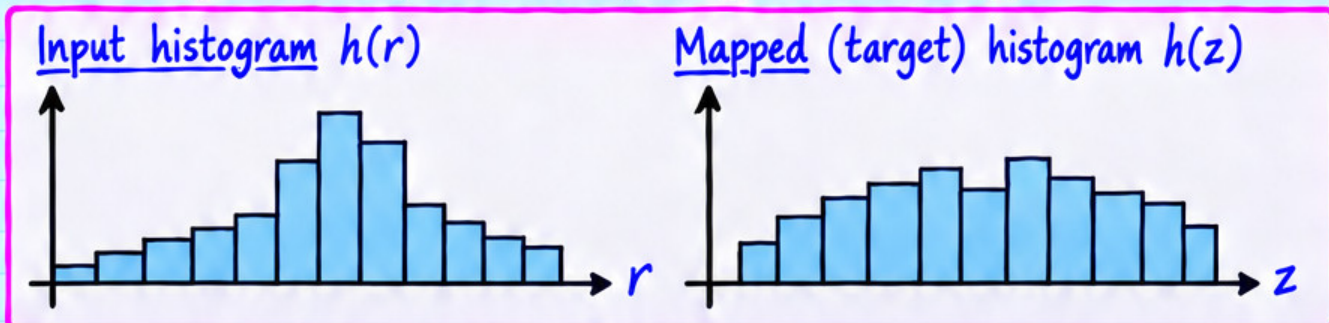
① Non-uniform quantization

- Choose uneven input bins: narrow where precision matters, wider where errors matter less.
- This is a coding/measurement decision, and it requires thresholds or a codebook.



② Non-uniform tonal mapping

- A histogram-based intensity map is generally nonlinear: output intervals need not be evenly spaced.
- A desired target histogram can be specified instead of seeking broad equalization (histogram matching).



③ Keep the vocabulary clear

quantization: continuous values $\rightarrow$ codes
 $x \in \mathbb{R} \longrightarrow k \in \{0, 1, \dots, M-1\}$

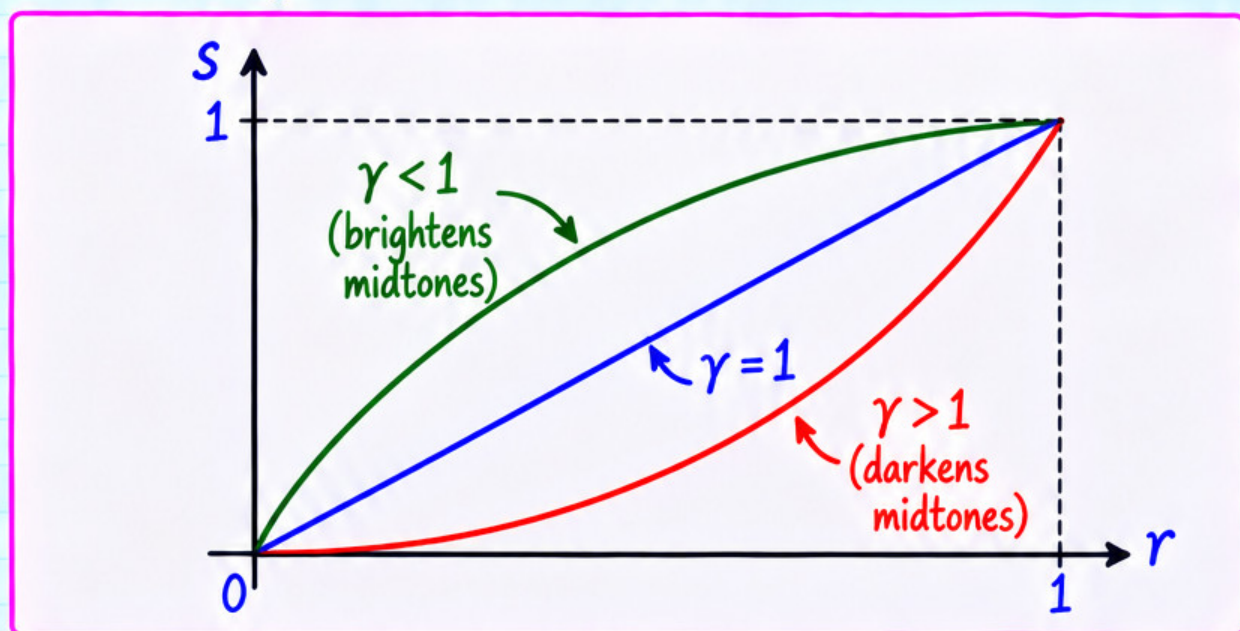
equalization: existing codes $\rightarrow$ remapped codes
 $k \in \{0, 1, \dots, M-1\} \longrightarrow k' \in \{0, 1, \dots, M-1\}$

Key takeaway: ● Neither process recovers detail that was lost earlier in sampling or coarse quantization.

Power-law tone transform vs. quantization precision

① Power-law (gamma) transformation

- $s = cr^\gamma$, for normalized intensity $0 \leq r \leq 1$.
- With $c=1$, $\gamma < 1$ brightens midtones; $\gamma > 1$ darkens them.
- This changes tone, not bit count.



② Quantization precision

- With k bits there are $L = 2^k$ levels.
- More bits usually give smaller rounding steps and less banding.

Ideal uniform rounding error:

$$|e| \leq \frac{\Delta}{2}$$

The term quantization power can also refer to error power; under a uniform-error model it is

$$\frac{\Delta^2}{12}$$

③ Do not mix them up

Key takeaway:

- Gamma is a point transform; bit depth controls the number of codes.
- State which idea your question means.

① Set target counts

- We have N pixels and L gray levels $(0, 1, \dots, L-1)$.
- Choose nonnegative integer target counts n_k for each level k , such that

$$\sum_{k=0}^{L-1} n_k = N, \quad n_k \geq 0 \text{ (integer)}$$

Example:

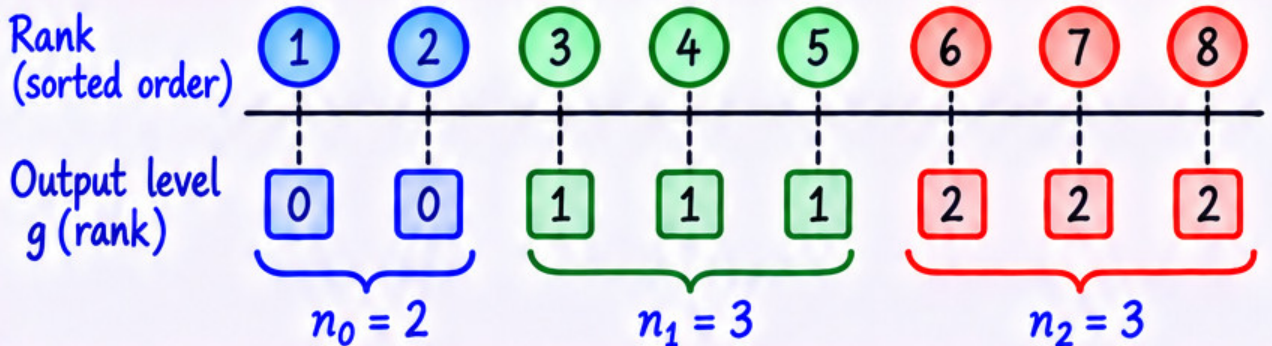
Exactly 8 pixels and 3 gray levels $(0, 1, 2)$.

Choose target counts $n = [2, 3, 3]$, i.e.,

$$n_0 = 2, \quad n_1 = 3, \quad n_2 = 3 \\ (2 + 3 + 3 = 8)$$

② Rank and assign

- Sort the 8 source pixels by intensity (ascending order).
- Break ties in a deterministic way (e.g., by original index).
- Assign output levels according to the target counts $n = [2, 3, 3]$.



- Equal source intensities (ties) may be mapped to different output levels, depending on their rank positions.

③ Difference from ordinary CDF matching

- Using the CDF lookup method, the actual output histogram counts may not match the target counts due to ties at some gray levels.
- The rank assignment method achieves the exact target counts (e.g., $[2, 3, 3]$ above), although the spatial appearance of the resulting image may vary depending on how ties are broken.

Key takeaway: ● Exact means the final output histogram counts match the target counts (e.g., $[2, 3, 3]$), not a unique image.

DIA: Global Histogram Equalization

Amol Adhava
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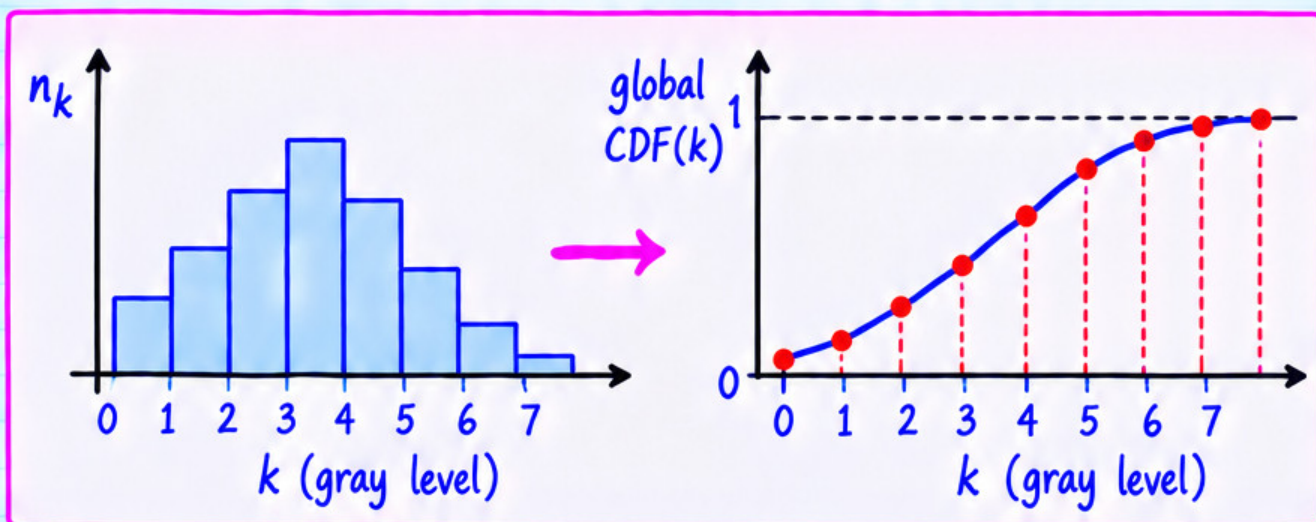
Chapter 3, section 3.3, begins at p. 148

① One mapping for the entire image

- All pixels having the same input level receive the same mapped output level.

$$\text{global CDF}(k) = \sum_{j=0}^k \frac{n_j}{MN}$$

$$s_k = \text{round} \left(\frac{(L-1) \times \text{global CDF}(k)}{1} \right)$$



② Strength and weakness

- A single lookup table is simple and fast, but a small low-contrast region may not receive enough local improvement.
- Uneven lighting and amplified noise can make global results undesirable for some images.

③ Revision contrast

- Local equalization uses neighborhood-dependent histograms; that is a later extension, not the same global mapping.

Key takeaway:

- Global = one CDF per image.
- Equalization changes contrast distribution, not underlying spatial detail.